Euclidean length $dx_1^2 + dx_2^2 + dx_3^2$), while the time carried by a clock ("proper" time) is given by $d\tau^2 = ds^2/c^2$. So, clearly, velocity through spacetime as just defined is given mathematically by $ds/d\tau$, which equals c.

8. Rudolf Carnap, "Autobiography," in The Philosophy of Rudolf Carnap, P. A. Schilpp, ed. (Chicago: Library of Living Philosophers, 1963), p. 37.

Chapter 6

1. Notice that the asymmetry being referred to—the arrow of time—arises from the order in which events take place *in* time. You could also wonder about asymmetries in time itself—for example, as we will see in later chapters, according to some cosmological theories time may have had a beginning but it may not have an end. These are distinct notions of temporal asymmetry, and our discussion here is focusing on the former. Even so, by the end of the chapter we will conclude that the temporal asymmetry of things in time relies on special conditions early on in the universe's history, and hence links the arrow of time to aspects of cosmology.

2. For the mathematically inclined reader, let me note more precisely what is meant by time-reversal symmetry and point out one intriguing exception whose significance for the issues we're discussing in this chapter has yet to be fully settled. The simplest notion of time-reversal symmetry is the statement that a set of laws of physics is time-reversal symmetric if given any solution to the equations, say S(t), then S(−t) is also a solution to the equations. For instance, in Newtonian mechanics, with forces that depend on particle positions, if $x(t) = (x_1(t), x_2(t), \ldots, x_{3n}(t))$ are the positions of n-particles in three space dimensions, then the fact that x(t) solves $d^2x(t)/dt^2 = F(x(t))$ implies that x(-t) is also a solution to Newton's equations, i.e. $d^2x(-t)/dt^2 = F(x(-t))$. Notice that x(-t) represents particle motion that passes through the same positions as $\boldsymbol{x}(\boldsymbol{t})$, but in reverse order, with reverse velocities.

More generally, a set of physical laws provides us with an algorithm for evolving an initial state of a physical system at time t_0 to some other time $t + t_0$. Concretely, this algorithm can be